

Mixture-of-Expert Blocks Contain Strong Hallucination Detection Signals

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Abstract

Despite their widespread use, Large Language Models (LLMs) remain limited by a fundamental problem: the generation of plausible but false content, known as *hallucinations*. Most existing detection methods operate at the answer or sentence level, yet per-token detection is essential for localizing hallucinated spans and enabling fine-grained interventions. In this paper, we explore the use of the Mixture-of-Experts (MoE) paradigm to address this gap. In MoE architectures, a single forward pass activates a sparse subset of experts (*i.e.*, distinct feedforward networks per layer) via a routing mechanism, producing internal signals (*e.g.*, router entropy, expert disagreement, and expert usage patterns) that are unavailable in dense architectures and have not been previously exploited for hallucination detection. To this end, we introduce *InnerExpert*, the first method to leverage these MoE-specific signals for per-token hallucination detection. *InnerExpert* combines routing-level and standard transformer signals into compact per-token feature vectors, classified by a lightweight detector trained on labels produced by an LLM-as-a-judge pipeline, which enables continuous model updates without manual annotation. Our results show that *InnerExpert* outperforms existing methods across five datasets and two MoE architectures, achieving up to 0.91 answer-level and 0.76 token-level AUROC, while requiring only a single forward pass.

Code — <https://github.com/joaopfonseca/InnerExpert-Hallucination-Detection>

1 Introduction

The reliability of Large Language Models (LLMs) is often questioned due to the risk of generating plausible but false content, commonly referred to as *hallucination* (Alansari and Luqman 2026). Hallucinations are particularly prevalent when LLMs are prompted for information that is outside their training distribution (*e.g.*, post-training events or highly specialized knowledge) (Vu et al. 2024). Despite the importance of this problem, detecting hallucinations reliably and efficiently remains an open challenge.

The Mixture-of-Experts (MoE) paradigm has become prevalent in frontier open-weight LLMs, as it allows for larger parameter counts and greater model performance at a feasible inference cost (Cai et al. 2025). A distinctive property of MoE models (as opposed to dense models) is that a

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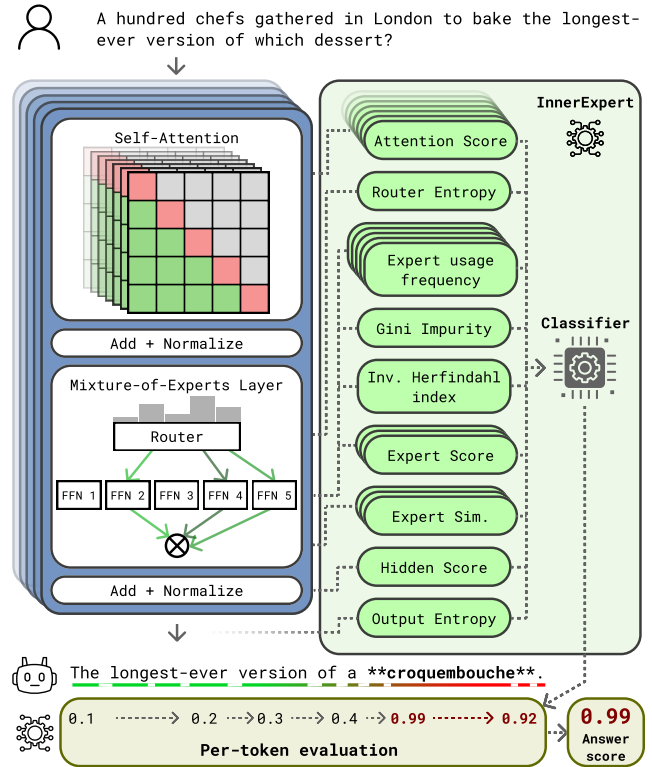


Figure 1: Overview of *InnerExpert*. A single forward pass through the MoE model exposes internal signals (*i.e.*, router distributions, per-expert hidden states, and expert usage patterns) which are combined into per-token hallucination scores.

forward pass selects, via a routing function, a sparse subset of *experts* (*i.e.*, feedforward networks) per layer (Fedus, Zoph, and Shazeer 2022). This process produces a large amount of signals, most of which are unavailable in dense architectures, such as router entropy, disagreement between experts’ hidden states, and the distribution of expert usage over a sequence. Figure 1 illustrates this approach at a high level.

Although this approach may allow the development of cheap and efficient hallucination detection methods, to the best of our knowledge, no prior work leverages MoE-specific

internal signals to detect hallucinations in LLMs. These signals have been shown to relate to epistemic uncertainty (Pavlitcka et al. 2025), providing a theoretical motivation for their use in hallucination detection; we develop this connection in Appendix A. Moreover, because MoE routing decisions are made at each token, these signals are naturally aligned with per-token granularity, enabling fine-grained localization of hallucinated spans. However, most existing methods operate at the answer or sentence level and do not provide such capability.

Existing hallucination detection methods can be grouped into three paradigms. Sampling-based approaches, such as SelfCheckGPT (Manakul, Liusie, and Gales 2023) and Semantic Uncertainty (Kuhn, Gal, and Farquhar 2023), probe consistency across samples (obtained via multiple forward-passes), which is expensive at inference time. INSIDE (Chen et al. 2024) is a sampling-based method that also utilizes hidden states over multiple responses. Internal-signal methods, such as LLM-Check (Sriramanan et al. 2024), and MIND (Su et al. 2024), operate on hidden states and attention in a single pass, but generally employ a limited subset of these signals and ignore the MoE-specific routing structure. Trainable detectors, such as HaluNet (Tong et al. 2025) and FacLens (Wang et al. 2025), learn classifiers using dense transformer signals over entire answers, but none of these operate at the token level. A complementary line of work on uncertainty estimation decomposes predictive uncertainty into aleatoric and epistemic components (Hüllermeier and Waegeman 2021; Yadkori et al. 2024), providing theoretical motivation for internal-signal approaches, while Pavlitcka et al. (2025) show that MoE-specific signals capture epistemic uncertainty in the vision domain. Additional details on the above approaches are provided in Appendix B. InnerExpert addresses these gaps by systematically extracting and combining routing-level signals with standard transformer signals into a unified hallucination detection framework.

InnerExpert combines the internal-signal and trainable-detector paradigms. It preserves the single-pass efficiency of internal-signal methods while adopting the trainable-classifier approach of detectors, leveraging the fact that MoE models expose additional routing-level signals, such as router entropy, expert disagreement, and expert usage patterns. To the best of our knowledge, no prior method explores these signals for hallucination detection in text generation. Furthermore, most existing methods operate at the answer or sentence level; per-token hallucination detection, which enables fine-grained localization of hallucinated spans, remains largely underexplored. **InnerExpert combines MoE-specific signals at per-token granularity, addressing both gaps simultaneously.** Importantly, we show that InnerExpert achieves state-of-the-art performance in hallucination detection using simple classifiers, *i.e.*, without employing complex architectures such as in HaluNet or FacLens, which we leave to future work. In summary, our contributions are as follows:

① We propose InnerExpert, a single-pass, per-token hallucination detector that observes the model’s internal states and leverages MoE-specific internal signals, requiring

no modifications to the host LLM or additional sampling.

② We formalize an unsupervised training approach for per-token hallucination detection, designed to train InnerExpert, based on LLM-based labeling of generated answers against reference evidence.

③ We demonstrate that InnerExpert achieves an average competitive performance of up to 0.91 AUROC at the answer-level and 0.76 AUROC at the token-level. in the hallucination detection task across five datasets and two different open-weight MoE architectures, against baselines spanning the sampling-based, internal-signal, and trainable paradigms.

④ We formalize an inventory of six MoE-specific signals alongside the standard hidden state and attention scores used in prior work, enabling a systematic study of which routing signals carry hallucination-discriminative information.

2 Background

Hallucination in LLMs. We consider an autoregressive LLM with parameters θ that, given a prompt (*i.e.*, sequence of input tokens) $\mathbf{x} = (x_1, \dots, x_U)$, generates a sequence $\mathbf{y} = (y_1, \dots, y_T)$ by sampling each token from $p_\theta(y_t | \mathbf{x}, y_{<t})$. A *hallucination* corresponds to a subset of fluent and internally coherent tokens $\mathbf{y}_h \subseteq \mathbf{y}$ that are factually incorrect or unsupported (Alansari and Luqman 2026).¹

Mixture-of-Experts and Its Internal Signals. A MoE replaces each feedforward sublayer of a standard transformer architecture with a sparse mixture of N expert networks $\{E_1, \dots, E_N\}$ and a gating (router) network, parameterized by $\mathbf{W}_l^g$, that selects a subset of k experts per layer, per token (Fedus, Zoph, and Shazeer 2022). Let $\mathbf{h}_{t,l} \in \mathbb{R}^d$ denote the hidden state at token position t entering layer l . Within layer l , the self-attention sublayer and subsequent post-attention layernorm produce an intermediate representation $\mathbf{h}'_{t,l} \in \mathbb{R}^d$ that serves as input to the MoE sublayer. The router produces a probability distribution over experts from this representation,

$$g_l(\mathbf{h}'_{t,l}) = \text{softmax}(\mathbf{W}_l^g \mathbf{h}'_{t,l}) \in \mathbb{R}^N, \quad (1)$$

a subset $\mathcal{S}_{t,l} \subseteq \{1, \dots, N\}$ of k experts is then selected (typically by top- k), and the MoE sublayer contribution is the weighted combination of expert outputs, added to the residual stream to form the hidden state entering the next layer:

$$\begin{aligned} \mathbf{h}_{t,l+1} &= \mathbf{h}_{t,l} + \text{MoE}_l(\mathbf{h}'_{t,l}), \\ \text{MoE}_l(\mathbf{h}'_{t,l}) &= \sum_{i \in \mathcal{S}_{t,l}} g_{l,i}(\mathbf{h}'_{t,l}) E_i(\mathbf{h}'_{t,l}). \end{aligned} \quad (2)$$

A single forward pass of an MoE model therefore exposes internal signals that are unavailable in dense architectures.

¹Some incorrect outputs reflect the *faithful propagation* of erroneous content already present in the training data or in retrieved context, rather than a generation failure per se; both sampling-based and internal-signal detectors treat such cases as low-uncertainty by design. The temporal-out-of-distribution evaluation we employ (Section 4) mitigates this conflation, since incorrect answers to post-training questions are predominantly fabrications.

Hallucination detection. `InnerExpert` aims to produce a score $s(y_t) \in [0, 1]$ for each generated token y_t and, by aggregation, an answer-level score $S(\mathbf{y}) = \text{agg}(\{s(y_t)\}_{t=1}^T)$ that indicates the likelihood that y_t (or $\mathbf{y}$) is hallucinated. We compute $s(y_t)$ from the internal signals $\mathcal{I}(y_t)$ that the model exposes during a single forward pass: aggregate and per-expert hidden states, attention weights, output logits, cumulative routing distributions and expert activations. A token is flagged as hallucinated when $s(y_t)$ exceeds a threshold τ , which can be tuned to trade off precision for recall using standard methods; the answer-level prediction is obtained by aggregating per-token scores (e.g., by mean or max) and thresholding. The specific signals and scoring function used in `InnerExpert` are defined in Section 3.

3 InnerExpert

`InnerExpert` combines MoE-specific internal signals and combines them via a meta-expert (i.e., a trained classifier) into per-token hallucination scores (as summarized in Figure 1). Since the host model’s signals are very high-dimensional, the Expert is intentionally fed a compact set of scores derived from the internal signals, rather than the raw hidden states, attention matrices and router logits. Although it is not required to modify the host model and no additional generations are needed beyond the forward passes that produce the answer under evaluation, some practical implementation challenges must be considered to achieve the full observability of the host model, which we detail in Appendix D.

3.1 Architecture

During generation, `InnerExpert` intercepts the model’s forward pass at each decoding step and collects a comprehensive set of internal signals at every layer. We distinguish between *standard signals*, available in any transformer-based architecture, and *MoE-specific signals*, which arise from the routing structure and are absent in dense architectures. All signals are computed cumulatively over the generated prefix $y_{\leq t}$, yielding a per-token value at each generation step t .

Standard signals. We adapt two complementary signals from LLM-Check (Sriramanan et al. 2024), which proposes single-pass internal signals that correlate with hallucination in prior work. The hidden state score captures representational redundancy across the generated sequence, while the attention score captures the model’s focus patterns; both are computed cumulatively over the prefix, aligning naturally with per-token granularity:

Hidden state score. For each layer l , let $\mathbf{H}_{t,l} \in \mathbb{R}^{t \times d}$ denote the matrix of hidden states $\{\mathbf{h}_{1,l}, \dots, \mathbf{h}_{t,l}\}$ collected up to position t . Following Sriramanan et al. (2024), we compute the centered covariance $\Sigma_{t,l} = \mathbf{H}_{t,l}^\top \mathbf{J} \mathbf{H}_{t,l} + \alpha \mathbf{I}$, where $\mathbf{J} = \mathbf{I} - \frac{1}{d} \mathbf{1} \mathbf{1}^\top$ centering over the hidden dimension and $\alpha > 0$ is a regularization constant (we use $\alpha = 0.001$). The hidden state score is the mean log-singular-value of $\Sigma_{t,l}$:

$$\phi_{t,l}^{\text{hid}} = \frac{1}{t} \sum_{j=1}^t \log \sigma_j(\Sigma_{t,l}), \quad (3)$$

where $\sigma_j(\cdot)$ denotes the j -th singular value. Low values indicate low-rank (redundant) hidden representations, associated with confident generation (Sriramanan et al. 2024).

Attention score. Let $\mathbf{A}_{t,l,h} \in \mathbb{R}^{t \times t}$ be the attention matrix of head h at layer l up to position t . The attention score is the log-cumulative sum of the diagonal entries of the attention kernel:

$$\phi_{t,l,h}^{\text{att}} = \sum_{j=1}^t \log(\mathbf{A}_{t,l,h}[j, j] + \epsilon), \quad (4)$$

where ϵ is a small constant for numerical stability. The diagonal of the attention kernel equals its eigenvalues, so this quantity approximates the log-determinant of the attention similarity matrix (Sriramanan et al. 2024).

MoE-specific signals. The routing structure of MoE models exposes additional signals unavailable in dense architectures. We define six signals covering three complementary aspects of routing behavior: routing uncertainty (router entropy), expert-level representation quality (expert hidden score, expert similarity), and routing distribution dynamics (expert usage distribution, Gini impurity, inverse Herfindahl). This selection is grounded in the connection between MoE routing and epistemic uncertainty (Pavlitska et al. 2025), detailed in Appendix A:

Router entropy. The entropy of the gating distribution at each layer,

$$\phi_{t,l}^{\text{rout}} = H[g_l(\mathbf{h}'_{t,l})] = - \sum_{i=1}^N g_{l,i}(\mathbf{h}'_{t,l}) \log g_{l,i}(\mathbf{h}'_{t,l}), \quad (5)$$

is high when the router is uncertain about which expert should handle the token, suggesting the input falls in a region where the model’s knowledge may be insufficient.

Expert hidden score. Each selected expert $i \in \mathcal{S}_{t,l}$ produces an output $\mathbf{e}_{t,l,i} = E_i(\mathbf{h}'_{t,l})$. We compute the hidden state score (Eq. 3) independently for each expert’s output sequence and aggregate via the routing weights:

$$\phi_{t,l}^{\text{exp-hid}} = \sum_{i \in \mathcal{S}_{t,l}} g_{l,i}(\mathbf{h}'_{t,l}) \phi_{t,l,i}^{\text{hid}}, \quad (6)$$

where $\phi_{t,l,i}^{\text{hid}}$ is the hidden state score computed from the per-expert output sequence $\{\mathbf{e}_{1,l,i}, \dots, \mathbf{e}_{t,l,i}\}$. This measures whether the activated experts individually produce confident representations.

Expert similarity. Low pairwise similarity among the activated experts’ outputs indicates disagreement, analogous to ensemble disagreement, which has been shown to capture epistemic uncertainty (Pavlitska et al. 2025). We compute the routing-weighted average of pairwise cosine similarities:

$$\phi_{t,l}^{\text{sim}} = \sum_{i \in \mathcal{S}_{t,l}} \sum_{j \in \mathcal{S}_{t,l}} g_{l,i}(\mathbf{h}'_{t,l}) g_{l,j}(\mathbf{h}'_{t,l}) \cos(\mathbf{e}_{t,l,i}, \mathbf{e}_{t,l,j}). \quad (7)$$

with low values indicating that the activated experts produce inconsistent representations for the current token.

Expert usage distribution. We maintain a cumulative count of expert selections up to position t , weighted by routing

probabilities and normalized into a distribution:

$$\mathbf{u}_{t,l} = \frac{\sum_{t'=1}^t \mathbf{c}_{t',l}}{\left\| \sum_{t'=1}^t \mathbf{c}_{t',l} \right\|_1} \in \Delta^{N-1}, \quad (8)$$

where

$$c_{t',l,i} = \begin{cases} g_{t,i}(\mathbf{h}'_{t',l}) & \text{if } i \in \mathcal{S}_{t',l} \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

retains the routing probability for each selected expert and zeros out the rest. This distribution captures how routing has evolved over the generated sequence.

Gini impurity. The Gini impurity of the expert usage distribution,

$$\phi_{t,l}^{\text{gini}} = 1 - \sum_{i=1}^N u_{t,l,i}^2, \quad (10)$$

is high when many experts are selected equally often (scattered routing) and low when few experts dominate.

Inverse Herfindahl index. The inverse Herfindahl index, or effective number of experts,

$$\phi_{t,l}^{\text{herf}} = \left(\sum_{i=1}^N u_{t,l,i}^2 \right)^{-1}, \quad (11)$$

captures how many experts are effectively contributing to the generation. A value close to 1 indicates a single expert dominates; a value close to N indicates uniform usage.

These signals expose the model’s routing behavior and the consistency of the experts’ hidden states at the token level. A summary of the metrics computed and their dimensionalities (for a single forward pass) is available in Appendix C. To the best of our knowledge, no prior work leverages these signals for hallucination detection.

Feature Assembly. In order for the signals to be used as input to the hallucination expert, for each generated token y_t , the per-layer signals are flattened into a single feature vector in a fixed order,

$$\Phi_t = \{\phi_{t,l}^{\text{hid}}, \phi_{t,l}^{\text{att}}, \dots, \phi_{t,l,h}^{\text{att}}, \phi_{t,l}^{\text{rout}}, \phi_{t,l}^{\text{exp-hid}}, \phi_{t,l}^{\text{sim}}, \phi_{t,l}^{\text{gini}}, \phi_{t,l}^{\text{herf}}, \mathbf{u}_{t,l}\}. \quad (12)$$

Signals are collected only for generated tokens (not the prompt prefix). During classifier training, all features except the expert usage distribution (which is already bounded in $[0, 1]$) are standardized via z-score normalization; the expert usage ratios are passed through unscaled.

Per-token hallucination scoring. As stated in Section 2, our goal is to produce a scoring function $s(y_t, \Phi_t) \in [0, 1]$ that predicts the probability that a generated token y_t is hallucinated. In `InnerExpert`, s is a lightweight classifier trained on labeled data (Section 3.2). Any standard ML classifier can be used; we evaluate five families (Logistic Regression, Random Forest, XGBoost, a multilayer perceptron, and a Transformer encoder). Full hyperparameter grids are provided in Appendix D.

Answer-level aggregation. Finally, the answer-level score is obtained by aggregating per-token scores:

$$S(\mathbf{y}) = \text{agg}(\{s(\Phi_t)\}_{t=1}^T), \quad (13)$$

where `agg` is the mean or max operator. In our implementation we use the max operator, *i.e.*, we consider an answer to be hallucinated if any of its tokens is predicted to be hallucinated. Therefore, an answer is flagged as hallucinated when $S(\mathbf{y})$ exceeds a threshold τ . We discuss potential uses of τ in future work for domain certification and controlled text generation in Appendix G.

3.2 Training Supervision

We aim to set up a training regime for `InnerExpert` that does not require manual annotation of hallucinations and allows for continuous model updates without manual effort. To do this, we leverage the RealTime QA platform (Kasai et al. 2023) to obtain a stream of question-answer pairs $(\mathbf{x}, \mathbf{y})$ with reference evidence $\mathbf{e}$, where $\mathbf{y}$ is generated by the host model. In our experiments, we use questions posed between January 1st 2024 and December 31st 2025 for training. Afterwards, we form per-token hallucination labels with LLM-as-a-judge evaluation (Zheng et al. 2023). In our experiments, we use GLM-5.1 as the judge. For each question, the host model generates two answers: one with access to the reference evidence, to emulate retrieval-augmented generation, and one without, which is more likely to contain hallucinations.

Answer-level labels. For each generated answer, an LLM judge compares the answer against reference evidence and produces (i) a binary hallucination label and (ii) a list of hallucinated span strings (exact substrings from the answer). The judge is prompted to return structured JSON with a label field (1 = hallucinated, 0 = grounded) and a field listing the hallucinated substrings.

Token-level labels. The hallucinated spans are mapped to per-token binary labels: a token y_t is labeled as hallucinated ($\ell_t = 1$) if its character span overlaps with any hallucinated span, and as grounded ($\ell_t = 0$) otherwise. This produces a token-level supervision signal aligned with the feature vector Φ_t .

Train/validation split. The labeled data is split into training and validation sets stratified by hallucination rate, grouped by question identifier to prevent leakage between answers to the same question. The validation set is used to compute model performance while performing hyperparameter grid search on the classifier. The best-found hyperparameter set per model is used to train a final model on the combined training and validation sets. The final model is evaluated on a held-out test set, and the threshold τ is selected to maximize F1-Score over all the labeled data. Out-of-distribution evaluation is performed on data from a temporally disjoint period (see Section 4). We validate the quality of the LLM-as-judge labels against human annotations in Appendix F.2.

4 Experiments

Our experimental procedure is organized around four research questions (RQs): **RQ1:** Can `InnerExpert` serve

as a reliable indicator of per-token hallucination? **RQ2:** How does `InnerExpert` compare to existing hallucination detection baselines spanning the sampling-based, internal-signal, and trainable-detector paradigms? **RQ3:** Which signals contribute most to hallucination detection performance? **RQ4:** How reliably does the LLM-as-a-judge labeling pipeline produce ground-truth hallucination labels? The LLM-as-a-judge evaluation is detailed in Appendix F.2.

Baselines. We compare `InnerExpert` against baselines spanning the three paradigms identified in Section 1. These baselines are also briefly described in Appendix B. For sampling-based methods, we generate $K=5$ stochastic samples per question with temperature 0.7 and top- $p=0.9$.

- *Sampling-based.* We evaluate two variants of `SelfCheckGPT` (Manakul, Liusie, and Gales 2023): the NLI-based variant and a prompt-based variant. We also evaluate `Semantic Uncertainty` (Kuhn, Gal, and Farquhar 2023) and `Semantic Energy` (Ma et al. 2025).

- *Internal-signal.* We evaluate two scoring variants from `LLM-Check` (Sriramanan et al. 2024), attention score and hidden score, evaluated independently with per-token scores aggregated to answer level via mean and max. As standard floor baselines, we include the logit entropy (*i.e.*, per-token Shannon entropy of the model’s output distribution) and perplexity.

- *Trainable.* We evaluate `HaluNet` (Tong et al. 2025), a multi-branch neural detector that fuses token-level log-likelihoods, entropy scores, and hidden-state embeddings. `HaluNet` is trained with binary cross-entropy loss. As no open-source implementation of `HaluNet` is available, we provide a re-implementation based on the architecture described in the paper; full hyperparameters are provided in Appendix D.

Datasets. We train `InnerExpert` and all baselines exclusively on `RealtimeQA` (Kasai et al. 2023) data from January 2024 to December 2025, labeled via the unsupervised LLM-as-judge pipeline described in Section 3.2. We evaluate on two categories of test data:

1. A temporally out-of-distribution subset of `RealtimeQA` from January to June 2026 (365 questions), which tests whether the detector generalizes to post-training events.
2. Four additional QA datasets: `SQuAD` (Rajpurkar et al. 2016), `TruthfulQA` (Lin, Hilton, and Evans 2022), `NQ-Open` (Kwiatkowski et al. 2019; Lee, Chang, and Toutanova 2019), and `FreshQA` (Vu et al. 2024), sampled at 200 questions each, which test cross-dataset generalization to different question distributions.

All datasets contain English-language questions and answers. `RealtimeQA` and `SQuAD` additionally provide reference evidence; `TruthfulQA`, `NQ-Open`, and `FreshQA` do not. We leave multilingual hallucination detection to future work. Full dataset descriptions and summary statistics are provided in Appendix E.

Host models. We use two open-weight MoE models as host architectures: `OLMoE-1B-7B-0924-Instruct` (Muennighoff et al. 2025) (64 experts, 8 active per token) and `Gemma-4-26B-A4B-it` (Team et al. 2026) (128 experts plus 1 shared,

8 active per token). Although `Gemma 4` is a multimodal model, we use it only for text-to-text generation.

Generation settings. For each question, the host model generates answers using greedy decoding with a maximum of 65 new tokens. For datasets that provide reference evidence (`RealtimeQA` and `SQuAD`), we generate two answers per question: one without evidence (base) and one with retrieved evidence (RAG simulation). For datasets without evidence (`TruthfulQA`, `NQ-Open`, and `FreshQA`), we generate only the base answer. This procedure is applied identically during both training and evaluation. For sampling-based baselines, we generate stochastic samples using the settings described in “Baselines”.

Hyperparameters. The `InnerExpert` classifier is selected via grid search over five different classifiers: Logistic Regression, Random Forest, `XGBoost`, and a multilayer perceptron (MLP). Hyperparameters are scored by F1 on a single validation fold (10% of training data, grouped by question identifier), and the best-performing hyperparameter sets per classifier is refit on the combined training and validation data. The decision threshold τ is then F1-optimized on this combined set via a precision-recall curve sweep. The complete hyperparameter grids are provided in Appendix F.

Performance metrics. We evaluate all methods at both answer level and token level. For threshold-independent comparison, we report AUROC. For threshold-dependent comparison, in Appendices F.3, F.4 and F.5 we report F1-Scores. All metrics are computed using the labels produced by the LLM-as-judge pipeline described in Section 3.2.

Hardware and software. All experiments were conducted on a machine with an AMD EPYC 9555P 64-core processor (128 threads), 258 GiB RAM, and two NVIDIA RTX PRO 6000 Blackwell Max-Q Workstation Edition GPUs (96 GiB VRAM each), using Ubuntu 26.04 LTS. The `InnerExpert` classifier and baselines were trained using `scikit-learn` 1.8 and `PyTorch` 2.10. Hallucination labels were generated using `GLM-5.1` accessed via the `DeepInfra` API.

4.1 Per-token hallucination detection (RQ1)

Table 1 reports token-level AUROC for `InnerExpert` and all baselines that produce per-token scores, across five evaluation datasets and both host models.²

Predictive performance. All five `InnerExpert` classifier variants exceed the best single-signal baseline in average token-level AUROC on both host models, with the strongest variant reaching 0.762 on `OLMoE` (LR) and 0.753 on `Gemma` (`XGBoost`), improvements of 0.12 and 0.20 over the best baseline. Classifier is not stable across host models: `XGBoost` and MLP lead on `Gemma`, while LR and MLP perform best on `OLMoE`. Among single-signal baselines, logit entropy and the `LLM-Check` hidden state score are the strongest; the `LLM-Check` attention score yields AUROC

²Sampling-based methods (`SelfCheckGPT`, `Semantic Uncertainty`, `Semantic Energy`) and `HaluNet` operate at answer level by construction and are therefore excluded from the token-level comparison.

Table 1: Token-level results (AUROC) across datasets and host models.

	FQA	NQO	RTQA	SQuAD	TQA	Rank	Avg
Gemma-4-26B							
IE (LR)	0.705	0.635	0.759	0.785	0.740	3.0	0.725
IE (MLP)	0.710	0.662	0.756	0.806	0.715	2.6	0.730
IE (RF)	0.638	0.651	0.749	0.788	0.724	3.6	0.710
IE (Transf.)	0.656	0.619	0.716	0.760	0.722	4.6	0.694
IE (XGB)	0.711	0.678	0.805	0.802	0.767	1.2	0.753
LC (att.)	0.310	0.331	0.370	0.332	0.223	8.0	0.313
LC (hid.)	0.525	0.544	0.512	0.614	0.562	6.2	0.551
Entropy	0.523	0.540	0.514	0.571	0.559	6.8	0.541
OLMoE-1B-7B							
IE (LR)	0.732	0.737	0.830	0.787	0.726	2.0	0.762
IE (MLP)	0.740	0.741	0.840	0.742	0.704	2.0	0.754
IE (RF)	0.618	0.531	0.633	0.483	0.707	5.6	0.594
IE (Transf.)	0.610	0.568	0.718	0.420	0.777	5.0	0.619
IE (XGB)	0.739	0.728	0.810	0.682	0.719	2.8	0.736
LC (att.)	0.258	0.287	0.267	0.481	0.224	7.8	0.303
LC (hid.)	0.590	0.569	0.590	0.670	0.609	6.0	0.605
Entropy	0.636	0.621	0.653	0.650	0.634	4.8	0.639

below 0.5 on both models, as the original LLM-Check formulation produces a single cumulative score per answer and our per-token adaptation did not yield a reliable per-token indicator. On the temporally out-of-distribution RealTimeQA test set, InnerExpert achieves its highest per-dataset AUROC, suggesting the learned mapping does not overfit to the training period.

Cross-dataset generalization. InnerExpert maintains a consistent advantage over single-signal baselines across all five evaluation datasets, including the three out-of-distribution QA datasets (FreshQA, NQ-Open, TruthfulQA) that differ substantially from the RealTimeQA training distribution (SQuAD provides evidence in a similar way RealTimeQA does). Threshold-dependent results (F1) are reported in Appendix F.3.

4.2 Comparison to baselines (RQ2)

Table 2 reports answer-level AUROC for InnerExpert against all baselines.

Baselines. InnerExpert (XGB) achieves the best average AUROC on both host models, surpassing HaluNet, the strongest trainable baseline, despite using structurally different input features and a complex classifier architecture. Sampling-based methods exhibit the largest cross-model variance: Semantic Energy is competitive on OLMoE (0.866) but near-random on Gemma (0.475), and Semantic Uncertainty shows a similar instability; SelfCheckGPT (NLI) is the most stable sampling-based baseline. Among internal-signal baselines, the LLM-Check hidden state score achieves competitive results. Threshold-dependent results (F1) are reported in Appendix F.4.

Inference cost. Figure 2 shows, InnerExpert (XGB) reaches the Pareto frontier, exceeding the AUROC of the

Table 2: Answer-level results (AUROC) across datasets and host models.

	FQA	NQO	RTQA	SQuAD	TQA	Rank	Avg
Gemma-4-26B							
HaluNet	0.836	0.835	0.937	0.945	0.952	3.2	0.901
IE (LR)	0.847	0.790	0.927	0.933	0.963	3.8	0.892
IE (MLP)	0.866	0.780	0.939	0.943	0.952	3.0	0.896
IE (RF)	0.752	0.832	0.870	0.919	0.937	5.8	0.862
IE (Transf.)	0.813	0.854	0.902	0.883	0.947	5.0	0.880
IE (XGB)	0.843	0.865	0.938	0.940	0.974	2.2	0.912
LC (att.)	0.650	0.711	0.616	0.887	0.724	8.0	0.718
LC (hid.)	0.680	0.834	0.445	0.949	0.704	7.0	0.722
Entropy	0.436	0.472	0.701	0.716	0.695	11.0	0.604
Perplexity	0.600	0.612	0.562	0.595	0.669	10.8	0.608
SCGPT (NLI)	0.646	0.630	0.713	0.760	0.788	8.2	0.707
SCGPT (P)	0.570	0.557	0.551	0.551	0.473	12.2	0.540
SemEnergy	0.437	0.429	0.553	0.506	0.452	13.4	0.475
SemUncert	0.526	0.512	0.614	0.670	0.597	11.4	0.584
OLMoE-1B-7B							
HaluNet	0.804	0.842	0.953	0.881	0.778	4.6	0.851
IE (LR)	0.920	0.784	0.924	0.802	0.898	4.2	0.865
IE (MLP)	0.973	0.923	0.953	0.692	0.826	4.2	0.874
IE (RF)	0.790	0.821	0.565	0.714	0.776	8.6	0.733
IE (Transf.)	0.721	0.510	0.649	0.329	0.839	10.0	0.610
IE (XGB)	0.943	0.926	0.928	0.786	0.825	4.0	0.882
LC (att.)	0.631	0.737	0.614	0.805	0.385	9.2	0.634
LC (hid.)	0.856	0.854	0.721	0.813	0.838	5.0	0.817
Entropy	0.245	0.170	0.530	0.768	0.191	11.8	0.381
Perplexity	0.146	0.051	0.182	0.367	0.113	13.8	0.172
SCGPT (NLI)	0.851	0.685	0.812	0.706	0.869	7.0	0.784
SCGPT (P)	0.622	0.550	0.644	0.564	0.531	11.2	0.582
SemEnergy	0.866	0.805	0.910	0.840	0.907	3.6	0.866
SemUncert	0.768	0.713	0.803	0.688	0.877	7.8	0.770

strongest baselines while remaining single-pass, whereas sampling-based methods cluster in the high-cost region without a proportionate predictive performance gain. We provide a complete table with the inference times and peak memory usage in Appendix F.4, Table 14. Vanilla generation costs 1.15 s per 100 tokens on OLMoE and 3.83 s on Gemma. Standard internal-signal baselines add negligible overhead ($< 10\%$). Extracting the full set of MoE signals raises the cost to $\sim 2.5\times$ vanilla, and the InnerExpert classifier adds only ~ 0.7 s on top ($\sim 3\times$ vanilla overall), with negligible difference across classifier variants and only $\sim 3\%$ peak GPU memory overhead. HaluNet is substantially cheaper but underperforms InnerExpert in AUROC. Sampling-based methods are the most expensive category due to $K=5$ generation passes, with SelfCheckGPT (Prompt) requiring a second model instance that roughly doubles GPU memory.

4.3 Signal contribution analysis (RQ3)

Table 3 isolates the contribution of individual MoE-specific signals by reporting answer- and token-level AUROC for each signal independently, alongside the standard signals and the full InnerExpert classifier.

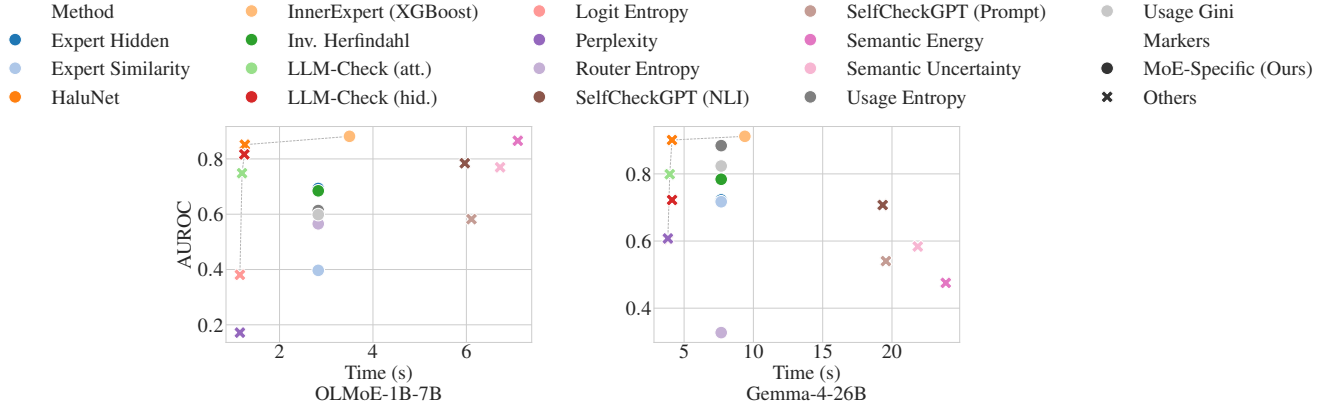


Figure 2: Inference time (per 100 tokens) vs. answer-level AUROC (averaged across five evaluation datasets) for each detection method, measured on 50 RealTimeQA questions per host model. The gray, dashed line represents the pareto frontier.

Table 3: Average AUROC for individual MoE signals, baselines, InnerExpert (XGB). Each value is averaged across all five evaluation datasets per host model. Answer-level scores use the best aggregation per method. Methods without a token-level counterpart are marked “—”.

Method	OLMoE-1B-7B		Gemma-4-26B	
	Answer	Token	Answer	Token
Inv. Herfindahl	0.685	0.675	0.784	0.602
Exp. Entropy	0.613	0.574	0.884	0.591
Exp. Gini	0.599	0.561	0.824	0.626
Expert Hidden	0.693	0.640	0.722	0.466
Expert Similarity	0.397	0.442	0.717	0.437
Router Entropy	0.565	0.574	0.327	0.446
Hid. Score	0.817	0.605	0.722	0.551
Att. Score	0.748	0.303	0.799	0.313
Logit Entropy	0.381	0.639	0.604	0.541
Perplexity	0.172	—	0.608	—
IE (XGB)	0.882	0.736	0.912	0.753
IE (MLP)	0.874	0.754	0.896	0.730
HaluNet	0.851	—	0.901	—
SCGPT (NLI)	0.784	—	0.707	—
SCGPT (Prompt)	0.582	—	0.540	—
SemUncert	0.770	—	0.584	—
SemEnergy	0.866	—	0.475	—

Individual MoE signals. No single MoE signal is uniformly strong across both host models. Expert entropy and usage Gini are the strongest individual MoE signals on Gemma, approaching HaluNet, but are considerably weaker on OLMoE. Conversely, expert hidden score is the strongest individual MoE signal on OLMoE but is mid-ranking on Gemma. Expert similarity is near-random on OLMoE yet moderately informative on Gemma, and router entropy is anti-correlated on Gemma.

Combination benefit. Combining all signals via the InnerExpert classifier yields consistent gains over the best individual signal, but the magnitude depends on the

host model. On OLMoE, where individual MoE signals are weak, the combination provides a significant improvement: IE (XGBoost) reaches 0.882 answer-level AUROC, a gain of 0.197 over the best individual MoE signal. On Gemma, where expert entropy alone already achieves 0.884, the combination adds a consistent margin (IE XGBoost: 0.912). At the token level, the signal combination yields a larger relative gain. These results indicate that the InnerExpert classifier extracts complementary information from the signal set that is not available to any individual signal.

Standard signals. Among the standard signals, the LLM-Check hidden score is strong on OLMoE (0.817 answer) but drops on Gemma (0.722), while the attention score shows the opposite pattern (0.748 vs. 0.799) but degrades to near-random at the token level. Logit entropy is anti-correlated at answer level on OLMoE but moderately informative at the token level (0.639). Threshold-dependent results (F1) are reported in Appendix F.5.

5 Conclusion

We presented InnerExpert, a single-pass, per-token hallucination detector that leverages MoE-specific internal signals, alongside standard transformer signals. Extensive experiments show that InnerExpert achieves up to 0.91 answer-level and 0.76 token-level AUROC, outperforming baselines spanning the sampling-based, internal-signal, and trainable-detector paradigms, while requiring only a single forward pass and modest computational overhead. Our signal contribution analysis reveals that no individual MoE signal is reliably strong across architectures, but combining them yields consistent gains, demonstrating that the combination of these signals yields information unavailable to any individual signal.

Limitations. Our human validation study (Appendix F.2) was evaluated on a small scale; we aim to expand it in future work with a larger sample size and annotator diversity. Additionally, our current evaluation is limited to the English language; future work will explore the generalizability of our

approach over a more diverse set of languages. Finally, this paper demonstrates that MoE-specific signals can be leveraged for hallucination detection. Hence, the classifiers used in InnerExpert are naturally lightweight. See appendix G for details.

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References

- Alansari, A.; and Luqman, H. 2026. Large language models hallucination: A comprehensive survey. *Computer Science Review*, 61: 100970.
- Azaria, A.; and Mitchell, T. 2023. The internal state of an LLM knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 967–976.
- Cai, W.; Jiang, J.; Wang, F.; Tang, J.; Kim, S.; and Huang, J. 2025. A survey on mixture of experts in large language models. *IEEE Transactions on Knowledge and Data Engineering*.
- Chen, C.; Liu, K.; Chen, Z.; Gu, Y.; Wu, Y.; Tao, M.; Fu, Z.; and Ye, J. 2024. INSIDE: LLMs’ Internal States Retain the Power of Hallucination Detection. In *The Twelfth International Conference on Learning Representations*.
- Chuang, Y.-S.; Xie, Y.; Luo, H.; Kim, Y.; Glass, J. R.; and He, P. 2024. DoLa: Decoding by Contrasting Layers Improves Factuality in Large Language Models. In Kim, B.; Yue, Y.; Chaudhuri, S.; Fragkiadaki, K.; Khan, M.; and Sun, Y., eds., *International Conference on Learning Representations*, volume 2024, 54158–54183.
- Emde, C.; Paren, A.; Arvind, P.; Kayser, M.; Rainforth, T.; Lukasiewicz, T.; Torr, P.; and Bibi, A. 2025. Shh, don’t say that! Domain Certification in LLMs. In *International Conference on Learning Representations*, volume 2025, 15394–15430.
- Farquhar, S.; Kossen, J.; Kuhn, L.; and Gal, Y. 2024. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017): 625–630.
- Fedus, W.; Zoph, B.; and Shazeer, N. 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120): 1–39.
- Fonseca, J.; Bell, A.; and Stoyanovich, J. 2025. SAFENUJUDGE: Safeguarding Large Language Models in Real-time with Tunable Safety-Performance Trade-offs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 19966–19980.
- Gal, Y. 2017. *Uncertainty in deep learning*. Ph.D. thesis, University of Cambridge.
- Han, J.; Kossen, J.; Razzak, M.; Schut, L.; Malik, S. A.; and Gal, Y. 2024. Semantic entropy probes: Robust and cheap hallucination detection in llms. In *ICML 2024 Workshop on Foundation Models in the Wild*.
- Hüllermeier, E.; and Waegeman, W. 2021. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine learning*, 110(3): 457–506.
- Ji, Z.; Lee, N.; Frieske, R.; Yu, T.; Su, D.; Xu, Y.; Ishii, E.; Bang, Y. J.; Madotto, A.; and Fung, P. 2023. Survey of Hallucination in Natural Language Generation. *ACM Comput. Surv.*, 55(12).
- Kasai, J.; Sakaguchi, K.; Le Bras, R.; Asai, A.; Yu, X.; Radev, D.; Smith, N. A.; Choi, Y.; Inui, K.; et al. 2023. Realtime qa: What’s the answer right now? *Advances in neural information processing systems*, 36: 49025–49043.
- Kuhn, L.; Gal, Y.; and Farquhar, S. 2023. Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. In *The Eleventh International Conference on Learning Representations*.
- Kwiatkowski, T.; Palomaki, J.; Redfield, O.; Collins, M.; Parikh, A.; Alberti, C.; Epstein, D.; Polosukhin, I.; Devlin, J.; Lee, K.; et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7: 453–466.
- Lahlou, S.; Jain, M.; Nekoei, H.; Butoi, V. I.; Bertin, P.; Rector-Brooks, J.; Korablyov, M.; and Bengio, Y. 2023. DEUP: Direct Epistemic Uncertainty Prediction. *Transactions on Machine Learning Research*.
- Lee, K.; Chang, M.-W.; and Toutanova, K. 2019. Latent retrieval for weakly supervised open domain question answering. In *Proceedings of the 57th annual meeting of the association for computational linguistics*, 6086–6096.
- Li, K.; Patel, O.; Viégas, F.; Pfister, H.; and Wattenberg, M. 2023. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 36: 41451–41530.
- Liang, T.; Du, Y.; Huang, J.; Kong, M.; Chen, L.; Li, Y.; Chen, S.; and Zhu, Q. 2025. MoLE: Decoding by Mixture of Layer Experts Alleviates Hallucination in Large Vision-Language Models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(18): 18684–18692.
- Lin, S.; Hilton, J.; and Evans, O. 2022. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers)*, 3214–3252.
- Ma, H.; Pan, J.; Liu, J.; Chen, Y.; Zhou, J. T.; Wang, G.; Hu, Q.; Wu, H.; Zhang, C.; and Wang, H. 2025. Semantic Energy: Detecting LLM Hallucination Beyond Entropy. arXiv:2508.14496.
- Manakul, P.; Liusie, A.; and Gales, M. 2023. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, 9004–9017.

- Muennighoff, N.; Soldaini, L.; Groeneveld, D.; Lo, K.; Morrison, J.; Min, S.; Shi, W.; Walsh, P.; Tafjord, O.; Lambert, N.; Gu, Y.; Arora, S.; Bhagia, A.; Schwenk, D.; Wadden, D.; Wettig, A.; Hui, B.; Dettmers, T.; Kiela, D.; Farhadi, A.; Smith, N.; Koh, P. W.; Singh, A.; and Hajishirzi, H. 2025. OLMoE: Open Mixture-of-Experts Language Models. In Yue, Y.; Garg, A.; Peng, N.; Sha, F.; and Yu, R., eds., *International Conference on Learning Representations*, volume 2025, 62061–62121.
- Pavlitcka, S.; Keskin, B.; Faßbender, A.; Hubschneider, C.; and Zöllner, J. M. 2025. Extracting uncertainty estimates from mixtures of experts for semantic segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 311–320.
- Rajpurkar, P.; Zhang, J.; Lopyrev, K.; and Liang, P. 2016. Squad: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 conference on empirical methods in natural language processing*, 2383–2392.
- Smith, F. B.; Kossen, J.; Trollope, E.; van der Wilk, M.; Foster, A.; and Rainforth, T. 2025. Rethinking Aleatoric and Epistemic Uncertainty. In *Forty-second International Conference on Machine Learning*.
- Sriramanan, G.; Bharti, S.; Sadasivan, V. S.; Saha, S.; Katakinda, P.; and Feizi, S. 2024. Llm-check: Investigating detection of hallucinations in large language models. *Advances in Neural Information Processing Systems*, 37: 34188–34216.
- Su, W.; Wang, C.; Ai, Q.; Hu, Y.; Wu, Z.; Zhou, Y.; and Liu, Y. 2024. Unsupervised real-time hallucination detection based on the internal states of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, 14379–14391.
- Team, G.; Abd, S. E.; Aggarwal, V.; Algayres, R.; Andreev, A.; Bachem, O.; Ballantyne, I.; Brick, C.; Cărbune, V.; Casbon, M.; et al. 2026. Gemma 4 technical report. *arXiv preprint arXiv:2607.02770*.
- Tong, C.; Zhang, Q.; Gao, J.; Jiang, L.; Liu, Y.; and Sun, N. 2025. HaluNet: Multi-Granular Uncertainty Modeling for Efficient Hallucination Detection in LLM Question Answering. *arXiv preprint arXiv:2512.24562*.
- Vu, T.; Iyyer, M.; Wang, X.; Constant, N.; Wei, J.; Wei, J.; Tar, C.; Sung, Y.-H.; Zhou, D.; Le, Q.; et al. 2024. Freshllms: Refreshing large language models with search engine augmentation. In *Findings of the Association for Computational Linguistics: ACL 2024*, 13697–13720.
- Wang, C.; Su, W.; Ai, Q.; and Liu, Y. 2026. Joint evaluation of answer and reasoning consistency for hallucination detection in large reasoning models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 33377–33385.
- Wang, Y.; Li, H.; Zou, H.; Zhang, J.; He, X.; Li, Q.; and Xu, K. 2025. FacLens: Transferable Probe for Foreseeing Non-Factuality in Fact-Seeking Question Answering of Large Language Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 18574–18593.
- Xia, Z.; Xu, J.; Zhang, Y.; and Liu, H. 2025. A survey of uncertainty estimation methods on large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, 21381–21396.
- Yadkori, Y. A.; Kuzborskij, I.; György, A.; and Szepesvari, C. 2024. To believe or not to believe your llm: Iterative prompting for estimating epistemic uncertainty. *Advances in Neural Information Processing Systems*, 37: 58077–58117.
- Zhang, S.; Yu, T.; and Feng, Y. 2024. Truthx: Alleviating hallucinations by editing large language models in truthful space. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 8908–8949.
- Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; et al. 2023. Judging LLM-as-a-judge with MT-bench and Chatbot Arena. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, 46595–46623.

A Hallucination Detection via Epistemic Uncertainty

Uncertainty in next-token prediction can be decomposed into *aleatoric* and *epistemic* uncertainty (Hüllermeier and Waegeman 2021). *Aleatoric* uncertainty reflects irreducible ambiguity in the data generating process (*e.g.*, multiple valid continuations of a prefix such as “The weather is . . .”), and is directly reflected in the entropy of the model’s output distribution. *Epistemic* uncertainty arises from the model’s lack of knowledge about the input region and is reducible with additional evidence. Existing hallucination detectors tend to operate on signals that primarily reflect aleatoric or aggregate uncertainty (*e.g.*, via multiple stochastic generations or logit-based predictions), leaving epistemic uncertainty largely under-explored as a detection signal.

The connection between epistemic uncertainty and hallucination provides a theoretical hook for detection: when epistemic uncertainty is high but the model emits a confident-looking answer (regardless of its aleatoric uncertainty), the output is likely to be unreliable and therefore hallucinated (Yadkori et al. 2024). This suggests that practical proxies for epistemic uncertainty can serve as useful hallucination detectors. The contributions in this paper allow defining useful epistemic uncertainty proxies for hallucination detection that are cheap to compute and available within a single forward pass, without requiring multiple stochastic generations or an external verifier. In the vision domain, MoE-specific signals have been shown to capture epistemic uncertainty, with expert disagreement acting as an analogue of ensemble disagreement (Pavlitska et al. 2025).

Uncertainty Decomposition. Following the Bayesian treatment of predictive uncertainty (Hüllermeier and Waegeman 2021), suppose the model parameters lie in a posterior $p(\theta)$. The marginal predictive distribution averages over this posterior,

$$p(y_t \mid \mathbf{x}, y_{<t}) = \mathbb{E}_{p(\theta)}[p(y_t \mid \mathbf{x}, y_{<t}, \theta)]. \quad (14)$$

The total predictive uncertainty, measured as the Shannon entropy of the marginal predictive, decomposes as

$$\underbrace{H[p(y_t \mid \mathbf{x}, y_{<t})]}_{\text{total}} = \underbrace{\mathbb{E}_{p(\theta)}[H[p(y_t \mid \mathbf{x}, y_{<t}, \theta)]]}_{\text{aleatoric}} + \underbrace{I(y_t; \theta \mid \mathbf{x}, y_{<t})}_{\text{epistemic}}, \quad (15)$$

where $I(\cdot; \cdot \mid \cdot)$ is the conditional mutual information between y_t and θ (Smith et al. 2025; Gal 2017). The *aleatoric* component captures irreducible ambiguity in the data-generating process (*e.g.*, multiple valid continuations of a prefix such as “The weather is . . .”) and is reflected in the entropy of a single conditional distribution $p(y_t \mid \mathbf{x}, y_{<t}, \theta)$. The *epistemic* component captures uncertainty about the parameters themselves and is reducible with additional evidence. Since $p(\theta)$ is not available in LLMs, neither term of (15) can be computed exactly; in practice, estimating uncertainty involves multiple model evaluations or stochastic generations, which are expensive at inference time. This is the case for methods like Semantic Uncertainty (Kuhn, Gal, and Farquhar 2023) and

SelfCheckGPT (Manakul, Liusie, and Gales 2023), which estimate an aggregate uncertainty measure that mixes aleatoric and epistemic.

Epistemic Uncertainty and Hallucination. In LLMs, the decomposition in (15) links epistemic uncertainty and hallucination (Yadkori et al. 2024): when epistemic uncertainty is high relative to aleatoric uncertainty, the model’s output is likely to be hallucinated. The mutual information can be expressed as an expected divergence among posterior samples,

$$I(y_t; \theta \mid \mathbf{x}, y_{<t}) = \mathbb{E}_{p(\theta)} \left[\text{KL}(p_\theta(y_t \mid \mathbf{x}, y_{<t}) \parallel p(y_t \mid \mathbf{x}, y_{<t})) \right], \quad (16)$$

where $\text{KL}(\cdot \parallel \cdot)$ is the Kullback-Leibler divergence. A token y_t is therefore likely to be hallucinated when this expected divergence is high relative to the aleatoric term, *i.e.*, when the model’s parameters, were they resampled, would diverge substantially yet the model nonetheless emits a confident-looking prediction. In the well-specified Bayesian limit this within-model disagreement tracks divergence from the data-generating distribution p_{data} , but this link is broken for misspecified models such as LLMs, whose parameters do not admit a posterior $p(\theta)$. We therefore treat MoE routing signals as empirical proxies of within-model disagreement, following the empirical results of Pavlitska et al. (2025) that expert disagreement captures epistemic uncertainty in vision models as the bridge. The challenge addressed in this paper is to define such proxies that are available within a single forward pass and do not require multiple stochastic generations or an external verifier.

B Related Work

We review prior work along the three paradigms introduced in Section 1: sampling-based detection, internal-signal detection, and trainable detectors. We then discuss uncertainty estimation methods, which provide the theoretical motivation for InnerExpert, and conclude with the emerging literature on MoE-specific signals, which constitutes the gap our work addresses.

Sampling-based approaches probe the consistency of a model’s outputs across multiple stochastic generations. Manakul, Liusie, and Gales (2023) introduce SelfCheckGPT, which detects hallucinations by measuring agreement among several sampled responses to the same prompt, either via an LLM-based checker or through token-level n-gram consistency. In this work, we use the two best-performing SelfCheckGPT variants for comparison: an NLI-based variant that detects contradictions between the original response and sampled responses using a DeBERTa model fine-tuned on MNLI, and a prompt-based variant that uses an LLM to assess whether each response sentence is supported by the sampled responses. Kuhn, Gal, and Farquhar (2023) propose Semantic Uncertainty, which clusters semantically equivalent samples and computes entropy over the resulting clusters, providing a more robust uncertainty estimate than raw token-level entropy. Farquhar et al. (2024) scale this approach and demonstrate that semantic entropy is effective for hallucination detection across long-form generation tasks. Ma

et al. (2025) extend this line of work with Semantic Energy, which replaces the probability-space entropy of Semantic Uncertainty with a Boltzmann-inspired energy function over semantic clusters, capturing model confidence even when sampled responses are semantically identical. Chen et al. (2024) introduce INSIDE, which measures the semantic consistency of multiple sampled responses in the model’s internal embedding space. Their EigenScore computes the log-determinant of the covariance matrix of sentence embeddings across K sampled responses, capturing semantic divergence in the dense representation space. While this approach retains more semantic information than text-level consistency metrics, it still requires multiple stochastic generations per prompt. Han et al. (2024) introduce Semantic Entropy Probes, lightweight probes trained to approximate semantic entropy from internal states, bridging sampling-based and internal-signal approaches. While these methods are effective, they require multiple forward passes per prompt, making them prohibitively expensive at inference time. InnerExpert, by contrast, operates on signals extracted from a single forward pass and requires no additional generations.

Internal signal-based methods leverage internal model states (e.g., hidden representations, attention patterns, or output distributions) collected during a single forward pass. Azaria and Mitchell (2023) show that a classifier trained on hidden states can distinguish factual from non-factual statements, demonstrating that hallucination-relevant information is encoded in internal representations. Sriramanan et al. (2024) propose LLM-Check, which compute hallucination scores from hidden states, attention kernels, and perplexity. These methods share InnerExpert’s single-pass efficiency, but they operate exclusively on individual signals available in any transformer architecture (hidden states, attention matrices, and output logits) and do not exploit the routing structure unique to MoE models. InnerExpert extends this paradigm by introducing MoE-specific signals (e.g., router entropy, expert disagreement, expert usage distributions) that are absent in dense architectures while combining them into a single hallucination score.

A related but distinct line of work modifies internal representations at inference time to *mitigate* hallucinations rather than detect them. Decoding by Contrasting Layers (Chuang et al. 2024) contrasts logits across layers to amplify factual knowledge localized in specific transformer layers. Inference-Time Intervention (Li et al. 2023) shifts activations along truth-correlated directions identified in a sparse set of attention heads. TruthX (Zhang, Yu, and Feng 2024) edits hidden representations in a learned truthful space to improve output factuality. MoLE (Liang et al. 2025) extends the layer-contrasting idea to vision-language models, dynamically selecting transformer layers as “experts” (Final, Second Opinion, and Prompt Retention) via a heuristic gating mechanism. We note that MoLE’s use of “expert” refers to transformer layers, not the routed feedforward networks of MoE architectures. These methods are complementary to hallucination detection but fall outside the scope of this paper, which focuses on detection rather than mitigation.

Trainable detectors over internal signals typically use lightweight classifiers over internal signals to produce hallucination scores. Tong et al. (2025) introduce HaluNet, which models hallucination risk using a multi-branch architecture that fuses token-level log-likelihoods, entropy scores, and hidden-state embeddings. Wang et al. (2026) propose a joint evaluation framework that assesses both answer and reasoning consistency for hallucination detection in reasoning models. Wang et al. (2025) present FacLens, a transferable probe for non-factuality prediction that analyzes how factual knowledge is embedded in hidden representations of fact-seeking questions, with the goal of cross-model transferability. Su et al. (2024) present MIND, a framework that trains an MLP on hidden states collected during generation, using an automated Wikipedia-based labeling scheme to avoid manual annotation. InnerExpert follows this trainable-detector paradigm but differs in two key respects: (i) it enriches the feature set with MoE-specific routing signals unavailable to prior methods, and (ii) it employs an unsupervised labeling pipeline based on LLM-as-judge evaluation against reference evidence, requiring no human annotations.

Uncertainty estimation in LLMs. Uncertainty estimation provides the theoretical motivation for InnerExpert, developed in Appendix A. Generally, for Machine Learning, Hüllermeier and Waegeman (2021) provide a formalization of aleatoric uncertainty (irreducible ambiguity in the data-generating process) and epistemic uncertainty (uncertainty arising from the model’s lack of knowledge), which is reducible with additional evidence. Lahlou et al. (2023) propose DEUP, a framework for directly predicting epistemic uncertainty by learning to estimate excess risk and subtracting an estimate of aleatoric uncertainty. Smith et al. (2025) revisit the aleatoric/epistemic decomposition and highlight practical challenges in disentangling the two components in deep learning models. Specifically for LLMs, Yadkori et al. (2024) introduce an iterative prompting approach for estimating epistemic uncertainty in LLMs via iterative prompting based on previous responses. Xia et al. (2025) provide a comprehensive survey of uncertainty estimation methods for LLMs, covering both sampling-based and internal-signal approaches.

Signals from Mixture-of-Experts The MoE architecture (Fedus, Zoph, and Shazeer 2022) introduces a routing mechanism that selects a sparse subset of expert networks per token, producing signals (router distributions, per-expert hidden states and expert usage patterns) that are not observable in dense models. Despite the growing prominence of MoE models in the LLM landscape, the information encoded in these signals remains largely unexplored for hallucination detection. Pavlitska et al. (2025) demonstrate that uncertainty estimates can be extracted from MoE models in the vision domain, showing that expert disagreement serves as an analogue of ensemble disagreement and captures epistemic uncertainty in semantic segmentation tasks. To the best of our knowledge, no prior work leverages MoE-specific internal signals for hallucination detection in LLMs.

C Additional Method Details

This appendix provides a complete reference for the signals used by `InnerExpert` (Table 4), the inference procedure (Algorithm 1), and the training pipeline (Algorithm 2).

C.1 Signal Summary

Table 4 lists all signals extracted by `InnerExpert`, their notation, category, and per-token dimensionality. Signals are grouped into *standard* signals (available in any transformer) and *MoE-specific* signals (derived from the MoE block). All per-layer signals are computed cumulatively over the generated prefix $y_{\leq t}$, yielding a per-token value at each generation step t . The ratio between the total per-token dimensionality and the original dimensionality of the signals used from the host model underscores the compactness of the representation, *i.e.*, detecting hallucinations does not require the full internal state of the host model. For the two host models used in this paper, the aggregate feature dimensionalities are presented in Table 5.

Table 5: Feature dimensionality for each host model. L : number of layers, H : number of attention heads, N : number of experts, $|\Phi_t|$: per-token feature dimensionality, Params: total model parameters, Act.: number of active parameters per token.

Model	L	H	N	$ \Phi_t $	Params	Act.
OLMoE-1B-7B-0924-Instruct	16	16	64	1360	~7B	~1B
Gemma-4-26B-A4B-it	30	16	129	4470	~26B	~4B

For Gemma, $N=129$ comprises 128 routed experts plus 1 shared expert that is always active.

C.2 Inference Procedure

Algorithm 1 outlines the `InnerExpert` inference pipeline. Given a question $\mathbf{x}$ (and optional evidence $\mathbf{e}$), the host model generates an answer $\mathbf{y} = (y_1, \dots, y_T)$ via greedy decoding. During generation, internal signals are collected at each token position and assembled into the feature vector Φ_t (Eq. 12). The trained classifier $s(\cdot)$ produces a per-token hallucination score $s(\Phi_t) \in [0, 1]$. The answer-level score $S(\mathbf{y})$ is obtained by aggregating per-token scores via the max operator (Eq. 13).

C.3 Training Procedure

Algorithm 2 outlines the `InnerExpert` training pipeline. Training data is sourced from RealtimeQA (January 2024–December 2025), with answers generated under two conditions (base and RAG) per question. The LLM-as-a-judge produces answer-level hallucination labels and hallucinated span strings, which are mapped to token-level binary labels. Feature vectors are extracted for all generated tokens and assembled into a training matrix. A stratified train/validation split, grouped by question identifier, prevents leakage between answers to the same question. Grid search over five classifier families (Logistic Regression, Random Forest, XGBoost, MLP, and a Transformer encoder) is scored by F1 on

Algorithm 1: `InnerExpert` Inference

Require: Question $\mathbf{x}$, evidence $\mathbf{e}$ (optional), trained classifier s , host model θ

Ensure: Answer $\mathbf{y}$, per-token scores $\{s(\Phi_t)\}_{t=1}^T$, answer-level score $S(\mathbf{y})$

- 1: Generate answer $\mathbf{y} = (y_1, \dots, y_T)$ via greedy decoding on θ , collecting internal states at each step
- 2: **for** $t = 1$ **to** T **do**
- 3: **for** $l = 1$ **to** L **do**
- 4: Collect hidden states $\mathbf{H}_{t,l} = \{\mathbf{h}_{1,l}, \dots, \mathbf{h}_{t,l}\}$
- 5: Compute $\phi_{t,l}^{\text{hid}}$ ▷ Eq. 3
- 6: **for** $h = 1$ **to** H **do**
- 7: Compute $\phi_{t,l,h}^{\text{att}}$ ▷ Eq. 4
- 8: **end for**
- 9: Compute $\phi_{t,l}^{\text{rout}}$ ▷ Eq. 5
- 10: Compute $\phi_{t,l}^{\text{exp-hid}}$ ▷ Eq. 6
- 11: Compute $\phi_{t,l}^{\text{sim}}$ ▷ Eq. 7
- 12: Update $\mathbf{u}_{t,l}$ ▷ Eq. 8
- 13: Compute $\phi_{t,l}^{\text{gini}}, \phi_{t,l}^{\text{herf}}$ ▷ Eqs. 10, 11
- 14: **end for**
- 15: Assemble Φ_t ▷ Eq. 12
- 16: Compute $s(\Phi_t) \in [0, 1]$
- 17: **end for**
- 18: Aggregate $S(\mathbf{y}) = \max_{t=1}^T s(\Phi_t)$ ▷ Eq. 13
- 19: **return** $\mathbf{y}, \{s(\Phi_t)\}_{t=1}^T, S(\mathbf{y})$

the validation fold. The best model per family is refit on the combined train+validation data, and the decision threshold τ is F1-optimized on this combined set via a precision-recall curve sweep.

D Implementation Details

Expert hidden state extraction. Standard MoE implementations such as HuggingFace’s `OlmoSparseMoeBlock` and `Gemma4TextExperts` do not expose per-expert hidden states—they return only the routing-weighted combination of selected expert outputs. To collect the per-expert signals required by `InnerExpert` (e.g., expert hidden score, expert similarity), we replace (via monkey-patching) the `forward` method of each MoE block in the host model with a modified version that saves the pre-routing per-expert outputs to a `last_experts_hidden` attribute *before* the routing weights are applied. For Gemma, the host decoder layer flattens hidden states from (B, S, H) to $(B \times S, H)$ before calling the MoE block; we handle this by registering a pre-forward hook on the parent layer that captures the original batch and sequence dimensions, enabling the patched forward to recover the correct 4D output shape. The design is architecture-agnostic: adding a new MoE model requires only implementing a `forward_<model>` function.

Memory usage. The instrumentation adds moderate GPU memory overhead. Benchmarking on 50 RealtimeQA questions, peak GPU memory for OLMoE increases from 12.7 GB (vanilla generation) to 13.1 GB

Table 4: Summary of signals extracted by InnerExpert. Each signal is computed cumulatively over the generated prefix up to position t , yielding a per-token value. The total per-token feature dimensionality is the sum of all signal dimensions. L = number of MoE layers, H = number of attention heads, N = number of experts.

Signal	Notation	Category	Description	Per-token dim.
Hidden state score	$\phi_{t,l}^{\text{hid}}$	Standard	Mean log-singular-value of hidden state covariance	L
Attention score	$\phi_{t,l,h}^{\text{att}}$	Standard	Log-cumulative sum of attention kernel diagonal	LH
Router entropy	$\phi_{t,l}^{\text{rout}}$	MoE	Shannon entropy of gating distribution	L
Expert hidden score	$\phi_{t,l}^{\text{exp-hid}}$	MoE	Routing-weighted hidden score of selected experts	L
Expert similarity	$\phi_{t,l}^{\text{sim}}$	MoE	Routing-weighted pairwise cosine similarity of experts	L
Expert usage	$\mathbf{u}_{t,l}$	MoE	Cumulative routing-weighted usage distribution	LN
Gini impurity	$\phi_{t,l}^{\text{gini}}$	MoE	Gini impurity of usage distribution	L
Inverse Herfindahl	$\phi_{t,l}^{\text{herf}}$	MoE	Effective number of experts from usage distribution	L
Total per-token dimensionality				$5L+LH+LN$

with full InnerExpert instrumentation, an overhead of approximately 3%. For Gemma, the increase is from 46.8 GB to 48.1 GB (2.8%). The dominant memory cost of InnerExpert is the per-expert hidden state tensor, which has shape (B, S, k, d) per MoE layer per generation step. On resource-constrained environments with limited VRAM, these tensors must be moved to CPU between generation steps to free memory for subsequent inference calls. On our hardware (2×96 GB VRAM), tensors can be accumulated on GPU and transferred in a single batch after generation completes. During baseline fitting, unused keys (e.g., `expert_usage` at 28 GB padded for Gemma) are filtered during data loading, substantially reducing peak RAM, at the cost of additional CPU time for filtering.

Storage requirements. Total disk usage for the project is approximately 1.7 TB, largely due to raw batch files saved during generation. Pretrained models (host LLMs and baseline models combined) account for 119 GB, while trained InnerExpert detectors take up 311 MB. The training data (RealtimeQA 2024–2025) requires 0.9 TB of raw files, the temporal OOD test set (RealtimeQA 2026) adds 0.17 TB, and the OOS datasets contribute 0.48 TB.

HaluNet re-implementation As no open-source implementation of HaluNet (Tong et al. 2025) is available, we re-implement it from the architecture described in the paper. The model has three branches: a log-likelihood branch and an entropy branch, each consisting of mean pooling followed by a 2-layer MLP, and a hidden-state embedding branch with two 1D convolutional layers (kernel size 3, padding 1) with ReLU activations and adaptive average pooling. Branch outputs are fused via attention and projected to a single logit. The hidden dimension is 128, the maximum sequence length is 50 (zero-padded and truncated), and dropout is 0.5. The model is trained with Adam (learning rate 10^{-3}), binary cross-entropy loss, for 20 epochs with a batch size of 32. Training uses the same LLM-as-judge binary hallucination labels as InnerExpert. The decision threshold is F1-optimized on the validation set. The embedding dimension is model-dependent and determined from the data at training time. All other hyperparameters follow the defaults specified in the

original paper.

E Datasets Description

This section provides detailed descriptions of the five datasets used in our experiments. RealtimeQA serves as both training data and temporal out-of-distribution test set, while SQuAD, TruthfulQA, NQ-Open, and FreshQA are used for cross-dataset generalization evaluation. Summary statistics, including sample counts, hallucination rates, and average answer lengths per host model, are reported in Table 6.

RealtimeQA (Kasai et al. 2023) is a dynamic question-answering platform that releases questions about current real-world events on a regular basis. Each question is accompanied by reference evidence in the form of retrieved passages. We use questions from January 2024 to December 2025 as training data and questions from January to June 2026 (365 questions) as the temporally out-of-distribution test set, evaluating whether the detector generalizes to post-training events.

SQuAD (Rajpurkar et al. 2016) is a reading comprehension dataset comprising over 100,000 questions posed by crowdworkers on Wikipedia articles. Each question is paired with a context passage (reference evidence) from which the answer span is drawn. We use the validation split and randomly sample 200 questions.

TruthfulQA (Lin, Hilton, and Evans 2022) is a benchmark of 817 questions across 38 categories designed to test whether language models avoid mimicking human falsehoods and common misconceptions. The dataset does not provide reference evidence, relying on the model’s internal knowledge. We use the train split to randomly sample 200 questions and use the provided answer as evidence.

NQ-Open (Kwiatkowski et al. 2019; Lee, Chang, and Toutanova 2019) is an open-domain question-answering benchmark derived from Google’s Natural Questions by discarding the accompanying evidence documents and retaining only questions with short answers. The dataset does not provide reference evidence, as the open-domain setting requires

Algorithm 2: InnerExpert Training

Require: RealtimeQA questions $\{\mathbf{x}_i\}$ with evidence $\{\mathbf{e}_i\}$, host model θ , LLM judge J

Ensure: Trained classifier s , threshold τ

- 1: **Phase 1: Answer Generation & Labeling**
- 2: **for** each question $\mathbf{x}_i$ **do**
- 3: Generate base answer $\mathbf{y}_i^{\text{base}}$ (no evidence) via greedy decoding on θ
- 4: **if** evidence $\mathbf{e}_i$ available **then**
- 5: Generate RAG answer $\mathbf{y}_i^{\text{rag}}$ (with evidence) via greedy decoding on θ
- 6: **end if**
- 7: **end for**
- 8: **for** each generated answer $\mathbf{y}_i$ **do**
- 9: LLM judge J produces binary label ℓ_i and hallucinated spans
- 10: Map spans to token-level labels $\{\ell_t\}_{t=1}^{T_i}$
- 11: **end for**
- 12: **Phase 2: Feature Extraction**
- 13: **for** each generated answer $\mathbf{y}_i$ **do**
- 14: **for** $t = 1$ **to** T_i **do**
- 15: Extract Φ_t following Algorithm 1, steps 2–11
- 16: **end for**
- 17: **end for**
- 18: **Phase 3: Model Selection**
- 19: Split data into train/validation sets, grouped by question identifier
- 20: Grid search over classifier families {LR, RF, XGBoost, MLP, Transformer}, F1-scored on validation fold
- 21: Select best hyperparameters per family; refit on combined train+validation
- 22: Optimize threshold τ for F1 on combined set via PR curve sweep
- 23: **return** s, τ

systems to retrieve evidence independently. We use the validation split to randomly sample 200 questions and use the provided answer as evidence.

FreshQA (Vu et al. 2024) is a dynamic benchmark with questions requiring fast-changing world knowledge, including questions with false premises that need to be debunked. We use a snapshot from November 2025 to randomly sample 200 questions.

F Additional Experiment Details

Table 7 reports the hyperparameter grids used for each classifier family.

F.1 Training Dataset Analysis

The block below contains the prompt template used to generate answers from the host models. The prompts are designed to elicit responses from the host model with and without access to reference evidence.

Table 6: Dataset statistics for all evaluation datasets and host models. Labels are from the LLM-as-a-judge approach (GLM-5.1). “Ans.” is the average number of words in the generated answer.

Dataset	N	Base	Eviden.	Halluc.	Grounded	Halluc. %	Ans.
Gemma-4-26B							
SQuAD	400	200	200	172	228	43.0	17.1
TQA	399	199	200	194	205	48.6	19.2
NQO	400	200	200	233	167	58.2	16.0
FQA	400	200	200	226	174	56.5	13.3
RTQA	730	365	365	397	333	54.4	15.3
OLMoE-1B-7B							
SQuAD	400	200	200	309	91	77.2	43.4
TQA	400	200	200	381	19	95.2	46.9
NQO	400	200	200	381	19	95.2	43.3
FQA	400	200	200	378	22	94.5	37.8
RTQA	730	365	365	635	95	87.0	43.8

Table 7: Hyperparameter grids for each classifier family. Hyperparameters used for OLMoE are boldfaced, the ones used for Gemma are underlined.

Classifier	Hyperparameter	Grid
Logistic Regression	C	{0.01, <u>0.1</u> , 1.0, 10.0 }
Random Forest	n_estimators	{300, 500, 1000 }
	max_depth	{3, 6, 10 }
	min_samples_leaf	{ 1 , 5}
XGBoost	n_estimators	{300, 500, 1000 }
	max_depth	{3, 6, 10 }
	learning_rate	{0.001, 0.01, 0.1 }
MLP	hidden_layer_sizes	{(128), (256,), (512,), (1024,)}
	α	{ 10^{-4} , 10^{-3} , 10^{-2} , <u>10^{-1}</u> }
	learning_rate_init	{ 10^{-3} , 10^{-4} }
	early_stopping	{True, False }
	d_model	{ 256 , <u>512</u> }
Transformer	n_heads	{ 4 }
	n_transformer_layers	{1, 2, 3 }
	dropout	{ 0.1 , <u>0.3</u> }
	optimizer_lr	{ 10^{-3} , 10^{-4} }
	max_epochs	{ 500 }

Generation Prompt

System message:

You are a helpful assistant who provides accurate and very concise answers to questions about recent events. Today is {DATE}.

User message (base, no evidence):

{QUESTION}

User message (with evidence):

Evidence: {EVIDENCE}

Question: {QUESTION}

The block below contains the prompt template used to label the training dataset using the LLM-as-a-judge approach. The prompt is designed to elicit a binary classification of the model's answer as either hallucinated or grounded, along with the specific spans of text that are hallucinated.

Judge Prompt

System message:

Return concise, valid JSON only.

User message template:

Given a question, evidence, and model answer, decide whether the answer contains unsupported or hallucinated content. Return valid JSON only with this exact schema:

```
{
  "label": 1 or 0,
  "hallucinated_spans": [
    "exact substring from answer",
    ...
  ]
}
```

Rules:

- label = 1 means hallucinated/unsupported, label = 0 means grounded/correct.
- hallucinated_spans must be exact substrings from the answer.
- if label = 0, hallucinated_spans must be an empty list.

Question: {QUESTION}

Evidence: {EVIDENCE}

Answer: {ANSWER}

The following blocks contain training data examples of hallucinations and non-hallucinations, wherein the hallucinated spans (as assessed by the judge LLM) are highlighted in red.

Example 1 (2024/01/05)

Question: Who was named Director of the Year at this week's Palm Springs International Film Festival Awards?

Evidence: The "Barbie" director was presented with the Director of the Year award by her film's stars Margot Robbie and America Ferrera at the awards ceremony in Palm Springs, California. Other winners included actors Cillian Murphy, Emma Stone and Carey Mulligan.

OLMoE (no evidence)

HALLUCINATED

At this week's Palm Springs International Film Festival Awards, the Director of the Year award was presented to Alfonso Cuarón, a renowned Mexican film director, producer, and screenwriter. Cuarón is known for his work on films such as "Y Tu Mamá También," "Children of Men,"

OLMoE (with evidence)

HALLUCINATED

The Director of the Year award was presented to the "Barbie" film's director, Liesl Tommy, at the Palm Springs International Film Festival Awards in California. Other winners included actors Cillian Murphy, Emma Stone, and Carey Mulligan.

Gemma 4 (no evidence)

HALLUCINATED

No awards were presented at this week's Palm Springs International Film Festival, as the festival is currently ongoing (January 4-14, 2024). The awards ceremony is scheduled for January 14, 2024.

Gemma 4 (with evidence)

HALLUCINATED

Greta Gerwig.

Example 2 (2024/05/31)

Question: What is the name of the new restaurant that opened in Rehoboth Beach, Delaware?

Evidence: ∅

OLMoE (no evidence)

HALLUCINATED

As of May 31, 2024, there isn't any widely reported new restaurant opening in Rehoboth Beach, Delaware, at least not under the name you provided. Rehoboth Beach is a popular beach destination in Delaware, and many restaurants come and go. If you have more information about the restaurant you're

OLMoE (with evidence)

HALLUCINATED

As of May 31, 2024, there is no new restaurant that has officially opened in Rehoboth Beach, Delaware, according to available information. However, there are several restaurants in the area that have been operating for years, such as The Salty Dog, The Beachcomber, and The Reh

Gemma 4 (no evidence)

GROUNDING

I do not have information on a specific new restaurant opening in Rehoboth Beach, Delaware, as your request does not specify a particular name or recent opening date. Please provide more details if you are looking for a specific establishment.

Gemma 4 (with evidence)

GROUNDING

Please provide the evidence mentioned in your prompt, as it was not included. Without the text, I cannot identify the specific restaurant you are referring to.

Example 3 (2024/01/05)

Question: What percentage of US adults have received the latest Covid-19 vaccine?

Evidence: Just 19% of adults and 8% of children have gotten the latest Covid-19 vaccine, CDC data shows. This comes as respiratory virus activity is high and rising across the US.

OLMoE (no evidence)

HALLUCINATED

As of January 05, 2024, a significant percentage of US adults have received the latest COVID-19 vaccine. According to data from the Centers for Disease Control and Prevention (CDC), as of January 2, 2024, approximately 70.8% of adults in the United States have received at least one dose of a COVID

OLMoE (with evidence)

GROUNDING

19% of US adults have received the latest Covid-19 vaccine, according to CDC data.

Gemma 4 (no evidence)

HALLUCINATED

As of January 2024, there is no single official percentage for the "latest" COVID-19 vaccine across all US adults, as uptake varies by age group and specific booster type. However, CDC data indicates that roughly **22%** of all US adults have received an updated 202

Gemma 4 (with evidence)

GROUNDING

19%

Example 4 (2024/01/05)

Question: Who beat 16-year-old darts prodigy Luke Littler to become the PDC world champion?

Evidence: British teen Littler, nicknamed "The Nuke", defeated former world No.1s Cross and van Gerwen to become the youngest-ever world championship finalist, before losing to Humphries. The 28-year-old new world champ took five consecutive sets to win 7-4 at London's Alexandra Palace on Wednesday.

OLMoE (no evidence) HALLUCINATED

As of January 05, 2024, 16-year-old darts prodigy Luke Littler did not yet win the PDC (Professional Darts Corporation) world championship. The PDC world champion for the 2023 season was Peter Wright, who defeated Mark Webster in the final. Peter Wright is a 38-

OLMoE (with evidence) HALLUCINATED

Luke Littler, also known as "The Nuke," was defeated by another player in the PDC world championship. Chris "Welly" Humphries, a 32-year-old player, won the match 7-4 at London's Alexandra Palace on Wednesday.

Gemma 4 (no evidence) HALLUCINATED

No one has beaten Luke Littler for the PDC World Championship yet; the 2024 tournament is currently ongoing. Luke Littler is a finalist/competitor in the event, but the champion has not been crowned.

Gemma 4 (with evidence) GROUNDED

Luke Humphries.

Example 5 (2025/04/04)

Question: More than 240 people fell ill recently due to a norovirus outbreak on which luxury cruise ship?

Evidence: The outbreak aboard the Queen Mary 2 comes on the heels of the worst year for cruise ship-based gastrointestinal outbreaks in over a decade.

OLMoE (no evidence) HALLUCINATED

As of April 03, 2025, there has been no reported norovirus outbreak on a luxury cruise ship involving more than 240 people falling ill. The norovirus is a highly contagious virus that can cause gastroenteritis, which includes symptoms such as nausea, vomiting, diarrhea, and stomach cramps. It is common on

OLMoE (with evidence) HALLUCINATED

The luxury cruise ship involved in the norovirus outbreak that affected more than 240 people was the Queen Mary 2. This incident occurred in early 2023, as mentioned in your evidence.

Gemma 4 (no evidence) HALLUCINATED

The norovirus outbreak occurred on the **Celebrity Eclipse**.

Gemma 4 (with evidence) GROUNDED

Queen Mary 2

F.2 LLM-as-a-Judge Label Validation (RQ4)

The training supervision pipeline described in Section 3.2 relies on LLM-as-a-judge labels as ground truth. In this section, we validate the quality of these labels against human annotations and compare them with a simpler weak labeling heuristic based on lexical and semantic overlap metrics.

Weak labeling heuristic. Lexical and semantic overlap metrics (e.g., BLEU, ROUGE, and BERTScore) are commonly used to evaluate hallucination in LLM outputs (Ji et al. 2023; Alansari and Luqman 2026). However, both these statistical metrics often fail to assess the factuality and faithfulness of generated text and lack robustness in aligning with human judgments of hallucination (Ji et al. 2023; Alansari and Luqman 2026). We observed this limitation empirically; the separability between hallucinated and grounded answers remained low across all three metrics. We nonetheless construct a weak labeling heuristic by computing F1-optimal thresholds for each metric (estimated from base vs. RAG separability on the training data) and averaging the resulting binary labels into a weak hallucination label. The inadequacy of this heuristic motivates the LLM-as-a-judge approach described in Section 3.2, which we validate below.

Human validation study. To assess label quality, we drew a stratified sample of 200 answers from the RealtimeQA training data (January 2024–December 2025). The sampling design consists of a 2×2 contingency stratification: for each host model (OLMoE and Gemma), we sample

Table 8: Confusion matrix between human annotations and LLM-as-a-judge labels/Weak labeling on the 200-sample validation set (pooled across both host models, 25 per 2×2 contingency cell per model).

LLM-as-a-judge Human	Grounded	Hallucinated	Total
Grounded	93	20	113
Hallucinated	7	80	87
Total	100	100	200

Weak heuristic Human	Grounded	Hallucinated	Total
Grounded	66	47	113
Hallucinated	34	53	87
Total	100	100	200

25 answers from each cell of the weak-label $\times$ LLM-label contingency table (both agree-grounded, both agree-hallucinated, weak-hallucinated/LLM-grounded, and weak-grounded/LLM-hallucinated), yielding $25 \times 4 \times 2 = 200$ samples. A single human annotator evaluated each sample, labeling it as hallucinated (1) or grounded (0) based on the question, the reference evidence, and the generated answer. The annotator had no access to the labels produced by either the weak heuristic or the LLM judge, ensuring that the validation is not biased by either labeling approach.

Results. Table 8 reports the confusion matrices between human annotations and both labeling approaches, pooled across both host models. The LLM judge achieves substantially higher agreement with human annotations, with 27 misclassifications out of 200 (86.5 % accuracy), compared to 81 misclassifications for the weak heuristic (59.5 % accuracy). The LLM judge’s errors are also more balanced: 20 false hallucinated labels (grounded by human, but hallucinated by the LLM judge) and 7 false grounded labels, whereas the weak heuristic exhibits 47 false hallucinated and 34 false grounded labels. This confirms that the LLM-as-a-judge approach produces more reliable training labels than the metric-based heuristic.

Confidence distributions. Tables 9 and 10 report the distribution of LLM-judge labels across the weak heuristic’s confidence scores (the calibrated hallucination probability) for hallucinated and grounded labels, respectively. For hallucinated labels, the judge’s labels are heavily concentrated in the $[0.8, 1.0]$ bin (74.4 % for OLMoE, 82.0 % for Gemma), indicating a good level of agreement flagging hallucinations. For grounded labels, confidence is concentrated in the $[0.0, 0.2]$ bin (24.9 % for OLMoE, 75.2 % for Gemma) and the $[0.2, 0.4]$ bin (67.5 % for OLMoE, 12.8 % for Gemma), indicating high confidence (low hallucination probability) for grounded answers. The clear separation between the two distributions suggests the LLM judge is well calibrated and that ambiguous cases (confidence near 0.5) are relatively rare. However, high disagreement between the weak heuristic and the LLM judge is observed in some cases, which

highlights the cases where the weak heuristic fails to label hallucinations while asserting high confidence.

Table 9: Confidence distribution, based on the weak labeling approach, for hallucinated labels (label=1)

Confidence	OLMoE (%)	Gemma (%)
$[0.0, 0.2[$	36 (0.9 %)	220 (9.6 %)
$[0.2, 0.4[$	598 (15.5 %)	88 (3.8 %)
$[0.4, 0.6[$	230 (6.0 %)	33 (1.4 %)
$[0.6, 0.8[$	121 (3.1 %)	71 (3.1 %)
$[0.8, 1.0]$	2862 (74.4 %)	1877 (82.0 %)
Total	3847 (100 %)	2289 (100 %)

Table 10: Confidence distribution, based on the weak labeling approach, for grounded labels (label=0)

Confidence	OLMoE (%)	Gemma (%)
$[0.0, 0.2[$	196 (24.9 %)	1764 (75.2 %)
$[0.2, 0.4[$	531 (67.5 %)	301 (12.8 %)
$[0.4, 0.6[$	44 (5.6 %)	50 (2.1 %)
$[0.6, 0.8[$	6 (0.8 %)	42 (1.8 %)
$[0.8, 1.0]$	10 (1.3 %)	189 (8.1 %)
Total	787 (100 %)	2346 (100 %)

Label distribution. Table 11 reports the label distribution by host model and evidence condition. OLMoE exhibits a strong class imbalance, with 83.0 % of answers labeled as hallucinated overall (99.7 % for base answers without evidence, 66.2 % for answers generated with evidence). Gemma is more balanced, with 49.4 % hallucinated overall (85.4 % base, 13.3 % with evidence). The contrast between base and evidence conditions creates a natural label diversity that benefits classifier training, though the class imbalance for OLMoE should be considered when interpreting threshold-dependent metrics such as F1 (cf. Section 4).

Table 11: Label distribution by model and evidence condition

Condition	N	Halluc.	Grounded	Halluc. %
OLMoE-1B-7B				
Base	2318	2312	6	99.7 %
Evidence	2318	1535	783	66.2 %
Overall	4636	3847	789	83.0 %
Gemma-4-26B				
Base	2318	1980	338	85.4 %
Evidence	2318	309	2009	13.3 %
Overall	4636	2289	2347	49.4 %

F.3 Per-token hallucination detection (RQ1 - Cont.)

Table 12 reports token-level F1. The threshold-dependent results reveal greater variability across classifier families than the AUROC analysis (Section 4.1). IE (MLP) and IE (LR) remain the most consistent performers, with IE (MLP) achieving the highest average F1 on OLMoE and IE (LR) on Gemma. In contrast, IE (XGBoost), the strongest variant under AUROC, suffers a sharp F1 degradation on Gemma, indicating that its F1-optimized threshold does not generalize across datasets. LC (att.) produces zero F1 on RealTimeQA for both models, reflecting threshold instability rather than a lack of discriminative signal.

Table 12: Token-level results (F1) across datasets and host models.

	FQA	NQO	RTQA	SQuAD	TQA	Rank	Avg
Gemma-4-26B							
IE (LR)	0.648	0.629	0.751	0.705	0.730	2.6	0.693
IE (MLP)	0.646	0.625	0.681	0.718	0.706	4.0	0.675
IE (RF)	0.571	0.575	0.756	0.706	0.774	3.2	0.676
IE (Transf.)	0.586	0.544	0.726	0.653	0.671	5.2	0.636
IE (XGB)	0.175	0.116	0.782	0.139	0.168	6.6	0.276
LC (att.)	0.649	0.645	0.000	0.602	0.720	4.2	0.523
LC (hid.)	0.652	0.627	0.738	0.643	0.727	3.2	0.678
Entropy	0.332	0.381	0.307	0.426	0.462	7.0	0.382
OLMoE-1B-7B							
IE (LR)	0.733	0.781	0.784	0.714	0.760	2.6	0.754
IE (MLP)	0.783	0.779	0.794	0.688	0.825	1.6	0.774
IE (RF)	0.732	0.604	0.631	0.508	0.826	5.0	0.660
IE (Transf.)	0.741	0.595	0.657	0.542	0.819	4.8	0.671
IE (XGB)	0.629	0.565	0.753	0.555	0.564	6.6	0.613
LC (att.)	0.763	0.773	0.000	0.616	0.815	4.2	0.594
LC (hid.)	0.684	0.658	0.144	0.628	0.756	5.4	0.574
Entropy	0.645	0.672	0.603	0.595	0.721	5.8	0.647

F.4 Comparison to baselines (RQ2 - Cont.)

Table 13 reports answer-level F1. IE (MLP) achieves the highest average F1 on both host models, consistent with its strong AUROC performance. IE (XGBoost) again degrades sharply on Gemma, confirming the threshold sensitivity observed at the token level. On OLMoE, F1 values are inflated across most methods (many exceeding 0.94), likely due to the extreme label imbalance on this model (Table 11); consequently, the F1 ranking is less informative than AUROC. Sampling-based methods exhibit low F1 on Gemma, with SelfCheckGPT (Prompt) reaching only 0.129, but recover on OLMoE. The full inference benchmark table is provided in Table 14; see Section 4.2 for discussion.

F.5 Signal contribution analysis (RQ3 - Cont.)

Table 15 reports F1 for individual signals and Machine Learning classifiers. The combination benefit observed under AUROC is preserved for IE (MLP), which achieves the highest F1 on both models at both levels. However, IE (XGBoost) underperforms several individual signals at F1 on both

Table 13: Answer-level results (F1) across datasets and host models.

	FQA	NQO	RTQA	SQuAD	TQA	Rank	Avg
Gemma-4-26B							
HaluNet	0.778	0.769	0.849	0.825	0.897	3.2	0.824
IE (LR)	0.790	0.763	0.849	0.873	0.904	2.4	0.836
IE (MLP)	0.828	0.780	0.876	0.890	0.868	2.0	0.849
IE (RF)	0.689	0.679	0.809	0.837	0.912	4.6	0.785
IE (Transf.)	0.751	0.787	0.840	0.760	0.884	3.6	0.804
IE (XGB)	0.444	0.410	0.804	0.472	0.636	9.6	0.553
LC (att.)	0.722	0.736	0.705	0.601	0.654	7.0	0.684
LC (hid.)	0.722	0.736	0.705	0.710	0.654	6.6	0.705
Entropy	0.585	0.658	0.722	0.689	0.735	7.8	0.678
Perplexity	0.317	0.395	0.332	0.424	0.561	12.4	0.405
SCGPT (NLI)	0.354	0.383	0.401	0.306	0.183	12.6	0.325
SCGPT (Prompt)	0.221	0.174	0.133	0.077	0.037	14.0	0.129
SemEnergy	0.643	0.675	0.633	0.588	0.587	9.2	0.625
SemUncert	0.594	0.590	0.604	0.597	0.581	10.0	0.593
OLMoE-1B-7B							
HaluNet	0.963	0.975	0.947	0.884	0.966	5.2	0.947
IE (LR)	0.955	0.973	0.931	0.830	0.963	9.0	0.930
IE (MLP)	0.973	0.981	0.950	0.875	0.978	1.8	0.951
IE (RF)	0.969	0.952	0.847	0.864	0.977	8.1	0.922
IE (Transf.)	0.972	0.972	0.905	0.871	0.976	6.6	0.939
IE (XGB)	0.952	0.957	0.877	0.857	0.919	11.0	0.913
LC (att.)	0.972	0.976	0.930	0.872	0.976	4.7	0.945
LC (hid.)	0.973	0.687	0.929	0.780	0.976	7.7	0.869
Entropy	0.969	0.972	0.935	0.876	0.976	5.2	0.946
Perplexity	0.968	0.972	0.934	0.874	0.974	6.9	0.944
SCGPT (NLI)	0.738	0.674	0.716	0.449	0.318	13.0	0.579
SCGPT (Prompt)	0.366	0.290	0.326	0.154	0.066	14.0	0.241
SemEnergy	0.961	0.974	0.939	0.896	0.975	5.2	0.949
SemUncert	0.970	0.974	0.930	0.870	0.974	6.6	0.944

models, the opposite of the AUROC finding, again reflecting threshold sensitivity rather than reduced discriminative power.

G Future Work

This work opens several avenues for future research. First, taking inspiration from methods such as Emde et al. (2025), one could certify the τ -lucidity of a model (*i.e.*, the guarantee that the model will not hallucinate with probability $1 - \tau$ within a given domain), building on the threshold discussion in Section 3.1. Second, adapting approaches such as Fonseca, Bell, and Stoyanovich (2025) for hallucination detection, or extending InnerExpert as a Controlled Text Generation method, is a promising direction. Third, while the unsupervised ground truth annotation approach is effective, it remains prone to noisy labels; more robust labeling methods could be explored. Fourth, while our results support the use of lightweight classifiers, the threshold sensitivity of XGBoost (Section F.3) suggests that ad-hoc classifier architectures or threshold calibration techniques could improve robustness. Fifth, the unsupervised training pipeline naturally extends to online model monitoring, enabling detection of hallucination rate drift over time. Finally, transferring

Table 14: Inference time (per 100 tokens) and peak GPU memory for each detection method, measured on 50 Real-TimeQA questions per host model. InnerExpert times include MoE feature extraction (SVD-based hidden scores, attention scores, expert routing signals) as well as classifier inference.

Method	OLMoE-1B-7B		Gemma-4-26B	
	Time per 100 tok (s)	Peak GPU (GB)	Time per 100 tok (s)	Peak GPU (GB)
Vanilla	1.146	12.660	3.825	46.810
Logit Entropy	1.152	12.670	3.830	46.880
Perplexity	1.152	12.670	3.830	47.010
LC (att.)	1.198	12.680	3.965	46.880
LC (hid.)	1.246	12.670	4.133	46.830
HaluNet	1.257	12.680	4.128	47.050
Router Entropy	2.829	12.820	7.677	47.470
Exp. Hidden	2.829	12.820	7.677	47.470
Exp. Similarity	2.829	12.820	7.677	47.470
Exp. Entropy	2.829	12.820	7.677	47.470
Usage Gini	2.829	12.820	7.677	47.470
Inv. Herfindahl	2.829	12.820	7.677	47.470
IE (LR)	3.493	13.080	9.376	48.110
IE (MLP)	3.492	13.080	9.377	48.110
IE (RF)	3.537	13.080	9.409	48.110
IE (XGBoost)	3.498	13.080	9.384	48.110
IE (Transf.)	3.495	13.110	9.384	48.180
SemUncert	6.722	14.230	21.860	48.360
SemEnergy	7.101	14.230	23.883	48.360
SCGPT (NLI)	5.966	14.200	19.331	48.340
SCGPT (Prompt)	6.108	26.850	19.562	59.340

InnerExpert across domains, tasks, languages, and MoE architectures without retraining is an important direction for scalability.

Table 15: Average F1 for individual MoE signals, baselines, InnerExpert (XGB). Each value is averaged across all five evaluation datasets per host model. Answer-level scores use the best aggregation per method. Methods without a token-level counterpart are marked “—”.

Method	OLMoE-1B-7B		Gemma-4-26B	
	Answer	Token	Answer	Token
Inv. Herfindahl	0.940	0.731	0.697	0.619
Exp. Entropy	0.881	0.639	0.748	0.671
Exp. Gini	0.869	0.597	0.745	0.671
Expert Hidden	0.945	0.698	0.707	0.667
Expert Similarity	0.945	0.634	0.740	0.523
Router Entropy	0.857	0.731	0.571	0.667
Hid. Score	0.869	0.574	0.705	0.678
Att. Score	0.945	0.594	0.684	0.523
Logit Entropy	0.946	0.647	0.680	0.382
Perplexity	0.944	—	0.405	—
IE (XGB)	0.913	0.613	0.553	0.276
IE (MLP)	0.951	0.774	0.849	0.675
HaluNet	0.947	—	0.824	—
SCGPT (NLI)	0.579	—	0.325	—
SCGPT (Prompt)	0.241	—	0.129	—
SemUncertainty	0.944	—	0.593	—
SemEnergy	0.949	—	0.625	—